Economic Impacts of UC Berkeley's Innovation & Entrepreneurship Ecosystem on the City of Berkeley and Alameda County

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Executive Summary

> *"The entrepreneurial energy and top-tier research fuel each other, combining to drive the campus's ultimate goal of making a lasting difference in the world. It's a dynamic duo that keeps the campus pushing boundaries and shaping the future."* - Rich Lyons, UC Berkeley Chancellor

UC Berkeley is consistently ranked as one of the top universities in the world at generating innovation and creating venture-backed startups. It has a broad array of incubators, accelerators, and entrepreneurship programs, as well as its research capacity and other innovation initiatives from its Innovation and Entrepreneurship (I&E) ecosystem. Guided by the university's strategic vision, this ecosystem has created a strong local network effect between research, entrepreneurship, and industry. The impact of these programs, and the startups that they produce, help make the campus a strong contributor to the growth of the Berkeley and Bay Area economies.

UC Berkeley-affiliated startups support more than 500 high-wage jobs in the City of Berkeley, with an average salary of $216,000[1]. Each startup job located in the city also supports an additional 1.05 jobs locally, resulting in a total employment multiplier of 2.05[2]. In total, the activity of these 182 UC Berkeley startups generates more than $300 million in regional value added across Alameda County, and approximately $200 million in the City of Berkeley alone[3].

Qualitative data gathered through stakeholder interviews for this study also highlighted many positive impacts the campus I&E ecosystem has on the City of Berkeley:

- I&E resources draw founders looking to launch or grow their startups in Berkeley.
- Successful startups working in campus incubators often first look at office and lab space within the city when looking to expand.
- The operational needs of I&E startups generate steady economic activity for local businesses across a variety of industries.

This data also highlighted a few key growth opportunities for the city and UCB:

- Perception of public safety and street conditions contributes to some founders hesitating to start or renew commercial leases in Berkeley.

[1] Calculated exclusively from the impacts reported by surveyed firms, this figure represents the ratio of total employment generated (direct, indirect, and induced) to the initial direct jobs created.
[2] Derived from survey data, this metric is defined as the quotient of total Direct Labor Income divided by the number of direct jobs created. This survey-level value aligns with the midpoint between the lower and upper extrapolation estimates.
[3] Modeled on survey data from 22 verified startups and extrapolated to the 182 startups based in Berkeley, CA, using IMPLAN regional multipliers for employment, induced spending and supplier linkages in Alameda County. See METHODOLOGY section for further explanation.

- ➢ Increased coordination between the city and I&E program leaders could increase the philanthropic involvement of startups in the Berkeley community.
- ➢ The perceived complexity of Berkeley's regulations concerns founders; more accessible guidance and streamlined regulatory processes would be beneficial.
- ➢ Increasing the availability of appropriate short-term lease office and lab space would address a challenge faced by startups and make them more likely to remain in Berkeley.

The remainder of this report discusses these topics in depth, before concluding with considerations for future studies on this topic.

1. BACKGROUND

2025 Startup Rankings
Alumni entrepreneur who have raised venture capital in the last decade

University	Founder Count	Company Count
1. UC Berkeley	**1,804**	**1,650**
2. Stanford	1,519	1,380
3. Harvard	1,355	1,237
4. Univ. Of Pennsylvania	1,206	1,113
5. MIT	1,131	1,019

Source: Top 100 Universities ranked for entrepreneurs. Pitchbook. September 15, 2025. https://pitchbook.com/news/articles/pitchbook-university-rankings

UC Berkeley is known for innovation that impacts the world, **having the highest number of venture-backed startups during the last decade** (Pitchbook). Its mission of knowledge creation and public service increasingly includes translating research into societal and economic benefit. University-affiliated startups play a central role in this process by moving research from the lab to the market. This not only advances the university's public mission but also drives job creation and workforce development in the region. City of Berkeley's Mayor Adena Ishii has highlighted the city's unique position as an innovation nexus, noting that by connecting UC Berkeley, Berkeley Lab, and local talent, the city has supported startup success across sectors, including clean energy, biotechnology, artificial intelligence, and food engineering. She further pointed out that the Berkeley Startup Cluster has played a central role in linking campus-based innovation with the local economy and supporting sustained entrepreneurial growth.

The City of Berkeley is compact, with a population of approximately 120,000 residents (Census Bureau, 2024), yet it supports a highly concentrated and diverse innovation environment shaped by the presence of UC Berkeley. The city's business landscape includes more than 13,000 licensed businesses (City of Berkeley, 2025), including UC Berkeley-affiliated startups in multiple industries, such as biotech, clean-energy, and

computing. Through our surveys and interviews, we also found that there are even more early stage startups in Berkeley that do not yet have business licenses. However, the local economic impact of these startups on the city has not been documented prior to this report.

Therefore, the UC Berkeley Office of Innovation & Entrepreneurship (I&E) commissioned the Haas Institute for Business Innovation (IBI), which partnered with the City of Berkeley and the Berkeley Startup Cluster, to examine the local economic impact of UC Berkeley's innovation and entrepreneurship programs[4].

2. METHODOLOGY

This research was scoped to address four questions:

1. How UC Berkeley-affiliated startups interact with the city and regional economy
2. What factors influence their decisions to remain in Berkeley or relocate
3. How the city can better support university innovation while encouraging long-term retention
4. The broader impact of UC Berkeley's innovation and entrepreneurship (I&E) ecosystem on the city.

In scope were the jobs, payroll, local spending, taxes, and investment activity of UC Berkeley-affiliated startups with a city presence, measured through surveys and administrative data.

Out of scope were startup demographic analysis, broader macroeconomic changes, long-term startup lifecycle outcomes, housing displacement effects, environmental externalities, and impacts not attributable to Berkeley-connected ventures or local expenditures.

Economic impact refers to the measurable economic activity generated in the City of Berkeley by startups and firms connected to UC Berkeley's innovation and entrepreneurship (I&E) programs. This includes **direct effects,** such as jobs created, payroll, local vendor spending, commercial space occupancy, tax contributions, and revenue generated by these ventures. It also includes **indirect effects** (i.e. supply-chain spending with local businesses) and **induced effects** (employee spending in the local economy). Together, these effects represent the total contribution of UC Berkeley-affiliated ventures to employment, income, output, and tax revenues within the

[4] The Berkeley Startup Cluster is a partnership between the City, UC Berkeley, Berkeley Lab, Berkeley Chamber and Downtown Berkeley Association with a mission to make Berkeley a more vibrant, equitable and accessible place for startups to launch and grow.

local economy. Our study focuses on annual, replicable measurement of direct, indirect, and induced impacts within Berkeley.

Economic impacts were estimated using the IMPLAN input-output modeling software. IMPLAN's modeling framework is supported by its county-level tax and industry sector datasets to generate multipliers for each level of impact. It is considered to be the industry standard for estimating the direct, indirect, and induced effects of economic activity. See Appendices B-D for details on what was included in the IMPLAN analysis.

Our research approach integrated survey data, qualitative insights, and economic modeling to assess startup activity and economic contributions to the City of Berkeley. Data sources included university-affiliated initiatives, the City of Berkeley, startup surveys, and interviews with founders and program participants.

Entrepreneurial programs were identified through the UC Berkeley BEGIN directory. Each program was coded by lifecycle stage, participant eligibility, and program type. This study also identified and surveyed startups that are affiliated with UC Berkeley.

The City of Berkeley Office of Economic Development provided an initial list of approximately 400 innovation companies operating in Berkeley during Q4 2024. Each company was reviewed to confirm operational status, activity within Berkeley, and a clear connection to UC Berkeley. Affiliation was determined through founder ties to UC Berkeley, participation in university department programs, and engagement with the broader UC Berkeley I&E ecosystem. Additional outreach to I&E program leaders ensured completeness, resulting **in a final count of 182 startups.**

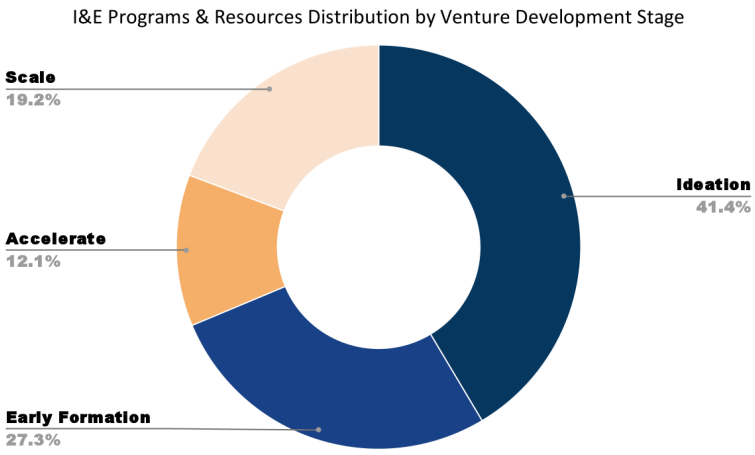
A 14-question survey collected data on employment, vendor spending, fundraising, and university and city connections. Responses were collected between October 29, 2025 and December 19, 2025. Thirty-four startups responded (22.2%), a voluntary sample that may not fully represent the full population but provided sufficient data to inform analysis. Interviews complemented quantitative findings and provided contextual insights.

Because Berkeley-affiliated startups generate economic value both within the city of Berkeley, CA, and across the broader Alameda County, impacts are reported separately for each geographic level to distinguish local impact from regional spillover effects. Operational impacts were modeled using Berkeley-based employment and spending, with multiplier effects estimated at the Alameda County level. Capital investment and tax impacts were modeled separately to maintain conservative, transparent reporting.

Regional impacts capture direct, indirect, and induced effects[5], as Berkeley-based activity propagates through supply chains, labor markets, and household consumption across Alameda County.

Because of highly skewed startup ecosystems (see Appendix B.2) and incomplete survey coverage of all startups, impacts are reported as a range based on two extrapolation scenarios that reflect different but plausible assumptions about ecosystem structure.

3. UC BERKELEY'S I&E ECOSYSTEM



UC Berkeley offers a large portfolio of programs supporting startups from ideation through early growth, spanning capability building, incubation, acceleration, and access to capital. The BEGIN directory[6] lists **100** entrepreneurial initiatives across campus that collectively cover the full startup journey. Together, these programs lower barriers for new founders and provide clear pathways for growth and commercialization through academic expertise, research infrastructure, industry partnerships, and funding. Below, we highlight **four examples** of UC Berkeley's I&E infrastructure that provide training, space, and capital to startups.

SCET: Preparing Students to be Founders & Early Validation

The Sutardja Center for Entrepreneurship & Technology (SCET) prepares students to move uncertain ideas from the classroom into the real-world, serving as an entry point to Berkeley's entrepreneurial ecosystem. Its model emphasizes interdisciplinary team formation, experiential learning, and capacity building before teams reach accelerator or financing stages. Rather than defining entrepreneurship by funding or company

[5] Direct effects represent employment and spending generated by startups within Berkeley. Indirect effects reflect economic activity created through supplier and service-provider purchases. Induced effects capture additional economic activity generated when employees spend their wages in the regional economy.
[6] https://begin.berkeley.edu/resources/

formation, SCET focuses on mindset and execution under uncertainty, prioritizing iteration, decision-making and learning through failure. As a result, outcomes are not measured solely by startups launched; many teams build prototypes and conclude at the course level, reflecting early-stage filtering. SCET's primary role is to broaden the pool of individuals capable of initiating entrepreneurial efforts, with continuation dependent on resources beyond the classroom.



BHEP: Helping teams transition into Venture-Focused Programs

Teams that continue beyond early validation often transition into venture-focused programs across campus.

The Berkeley Haas Entrepreneurship Program (BHEP) supports venture development through academic coursework, experiential learning, and engagement with industry and investor networks. Operating within the Haas Institute for Business Innovation, BHEP provides founders with structured training in business model development, market strategy, and capital planning. Its programs connect students and founders to faculty expertise, alumni entrepreneurs, and venture mentors, strengthening the pipeline of startups progressing toward acceleration and financing.

As ventures mature and require specialized facilities or additional validation, they may engage with campus infrastructure and acceleration programs.

Bakar Labs: Physical space

Bakar Labs provides a residency and shared facility system to help biotech startups access the scientific research and community resources of UC Berkeley. In FY24–25, Bakar Labs reported an annual budget of $4.6 million, with 15 full-time employees. It is estimated that 65% of its procurement expenditure occurs locally in Berkeley. Its tenant companies have placed 304 internships and/or jobs.[7]

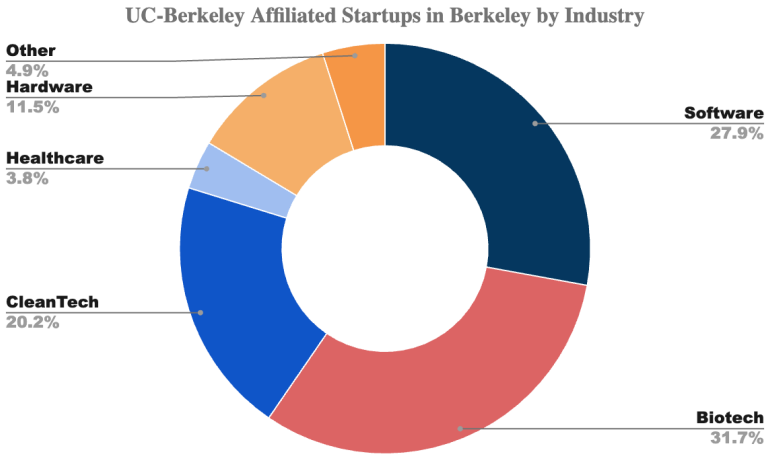


[7] Department survey.

Skydeck: Acceleration and Capital Support

Berkeley SkyDeck is one of the university's startup accelerators, operating through a public-private partnership model that integrates mentorship, investor access, and structured acceleration programming. In FY24–25, SkyDeck facilitated approximately 800 internships or job placements in Berkeley and reported annual supplier expenditures of about $1.5 million[8]. Skydeck provides both non-dilutive funding - such as sponsorships, grants and project-based funds - and dilutive funding. For example, SkyDeck's provided $3.2 million in sponsorship and philanthropic funds to subsidize early-stage operating costs without requiring equity. It also supports startups' market validation via industry, enterprise, and clinical partnerships.

The UC-Berkeley Affiliated Startups in Berkeley



Our review of the list of 400 startups, provided by the Berkeley Startup Cluster, confirmed that 148 companies have clear affiliations with UC Berkeley's I&E ecosystem. In addition, when surveys were distributed through university departments and entrepreneurship programs to reach affiliated startups directly, 34 additional startups were identified, bringing our total count to **182 UC Berkeley-affiliated startups that are based in the City of Berkeley**.These companies operate across a diverse set of industries, with the largest concentrations in biotech, software, and cleantech.

4. RESULTS - ECONOMIC IMPACT

Because of the data constraints mentioned in the methodology section of this report, we must emphasize that the results should be interpreted not as a precise census of ecosystem activity, but as a directional benchmark for strategic planning and policy discussion. (See Appendix E for the modeling assumptions used.)

[8] Department survey.

City of Berkeley – Direct Economic Activity[9]

The **22 startups we surveyed directly support 107 jobs in the city of Berkeley, generating approximately $23 million in annual payroll,** implying an average salary of roughly $216,000. (A job is defined as full-time equivalent employees working two or more days per week within city limits.) This compensation level reflects the high-skill, high-value nature of innovation economy employment associated with UC Berkeley–affiliated startups.

Survey responses reveal substantial heterogeneity in firm size and economic activity. This skewed structure is characteristic of startup ecosystems and motivates the distinction between anchor firms and long-tail firms, rather than reliance on simple averages. Detailed sample characteristics are provided in Appendix B.2.

Applying the extrapolation framework described in Section 4.3, direct employment within the City of Berkeley is estimated to range between 516 and 643 jobs, with corresponding annual payroll between $112.3 million and $138.5 million, and **direct value added between $203.6 million and $249.6 million.**

Table 1: Executive Summary of Economic Impacts

BERKELEY STARTUP ECOSYSTEM - IMPACT SUMMARY			
Metric	**Operational Survey impact**	**Lower Extrap.**	**Higher Extrap.**
A. CITY OF BERKELEY (Direct Activity)			
Employment	107[10]	516	643
Annual Payroll (Millions)	$23.20[11]	$112.27	$138.46
Value Added (Millions)	$42.04	$203.57	$249.55
B. REGIONAL FOOTPRINT (Alameda County - Total Impacts)			
Employment	220	1,061	1,319
Annual payroll (Millions)	$35.69	$172.38	$212.89
Value Added (Millions)	$63.22	$305.61	$375.70
Output (Millions)	$93.37	$454.01	$558.31

Source: IMPLAN Output and extrapolation.

[9] *NB: For Key definitions see Appendix D.*
[10] Full-time equivalent employees working ≥2 days/week in Berkeley.
[11] Total compensation to Berkeley-based workers.

Regional Economic Footprint – Alameda County

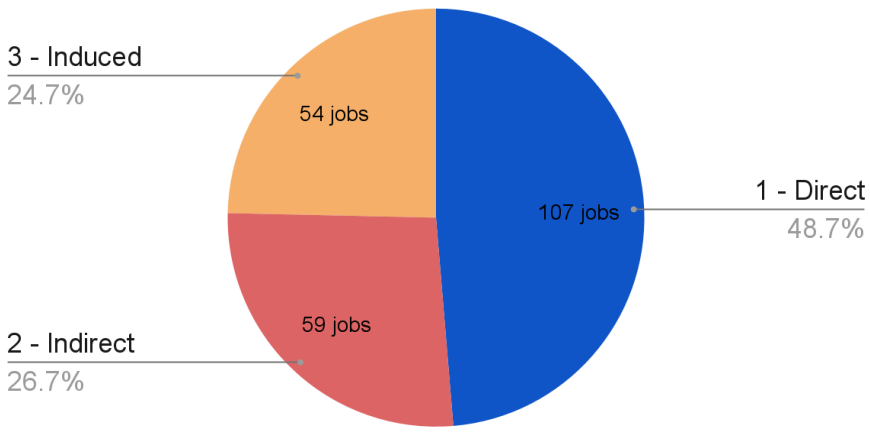
IMPLAN modeling estimates that the surveyed startups support 220 (direct, indirect and induced) regional jobs, of which roughly half (107/220) are physically located in Berkeley. Extrapolated results indicate total regional employment between 1,061 and 1,319 jobs, as summarised in Table 1.

The composition of these impacts is illustrated in the following breakdowns, with detailed numerical results provided in Appendix B.9.

This breakdown illustrates that approximately half of total regional employment is physically located in Berkeley, while the remainder *represents spillover effects across Alameda County.*

As Figure 3 illustrates, each job located in Berkeley supports an additional 1.05 jobs elsewhere in Alameda County, yielding a total employment multiplier of 2.05. **In practical terms, every UC Berkeley-affiliated innovation industry job ultimately supports just over two regional jobs once supply-chain and household spending effects are considered.**

Figure 3: Regional Employment Breakdown - Survey Sample (220 jobs)



The labor income multiplier (1.54) is lower than the employment multiplier because it reflects wage differences. In contrast, the value added (1.50) and output (1.51) multipliers are similar to the income multiplier, as they primarily capture the same supply-chain linkages and spending patterns. These multipliers reflect both the high-wage, knowledge-intensive structure of Berkeley's startup ecosystem and the integrated nature of the Alameda County regional economy.

These regional results are consistent with the direct Berkeley impacts reported above and reinforce the scalable and regionally integrated nature of the Berkeley startup ecosystem.

5. RESULTS - QUALITATIVE INSIGHTS

Our interviews with founders of UC Berkeley-affiliated startups illuminated consistent trends on the key positive impacts that both these startups and Berkeley's I&E programs have on the city of Berkeley's economy. The positive reactions provided context for the high rating of 8.8, that survey respondents gave Berkeley as a place to grow their startup[12]. This supports the results produced by the quantitative economic impact model and reaffirms that I&E activity at UC Berkeley directly materially impacts Berkeley's economy. Interviewees also gave suggestions that we grouped into three opportunities to increase the city of Berkeley's attractiveness to new startups and maximize the positive impact of existing startups on the Berkeley community.

I&E Impact on the Number of Startups in Berkeley



The **most direct impact of the I&E ecosystem is that startups are drawn to Berkeley as an attractive location to launch or grow**. Many founders mentioned that the launch of their companies would not have been possible without the breadth of resources provided by the I&E ecosystem. A notable example of this is AgriSolar. This social impact startup, founded by UC Berkeley grad students, uses a revenue-sharing model to provide affordable, solar-powered irrigation systems to farmers in Africa. Agrisolar initially received small grants from the Haas School of Business before winning the

[12] UC Berkeley startup survey question, "On a scale of 0-10, with 0 being the least likely and 10 being the most likely, how likely are you to recommend Berkeley as a place to launch or grow a startup?" 34 Respondents. Average rating: 8.8/10. Standard Deviation:1.4. % of respondents that rated Berkeley 10 of out 10: 48%.

grand prize at the Big Ideas Contest held on campus. The $20,000 award allowed them to expand their operations (see photo to the right) in Berkeley.

After receiving support from UC Berkeley to get their startups off the ground, **founders who have obtained sufficient success to move their startups off campus often first look at office and lab space within the city of Berkeley** because of their familiarity with the area and desire to continue leveraging the resources offered on campus. Azalea Therapeutics is one such example. Founded by Berkeley alum Jenny Hamilton, the Bakar Labs-based company engineers T cells to fight cancer. It has grown to 21 employees after raising $82 million in Series A funding. Azalea is now moving into a larger space at Foundry 31, keeping its operations in Berkeley. This path illustrates how UC Berkeley's support encourages growing startups to remain in the city and strengthen its biotech reputation.

Qualitative Impacts of Startups Within Berkeley

Startups that choose to locate in Berkeley boost the local economy in several ways. They create new, high-paying jobs, strengthen the local labor market, and increase employee spending at nearby businesses.

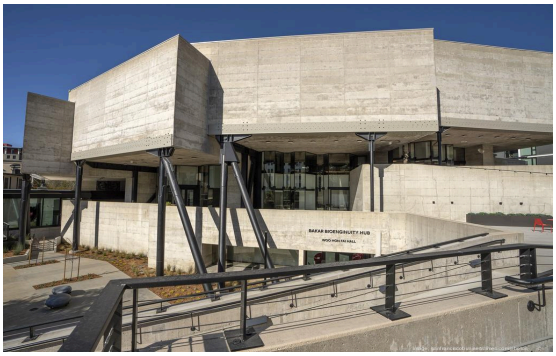
Day-to-day operations also support Berkeley vendors. Many campus-affiliated startups regularly buy food from local restaurants and rely on nearby suppliers for office products, printing, cleaning, and facilities services. These recurring purchases provide stable revenue for small and mid-sized businesses and reinforce local supply chains.

By leasing offices and laboratories in Berkeley, **startups increase demand for commercial real estate** and help sustain specialized facilities like Foundry 31 and Berkeley Commons. Their presence raises occupancy rates, supports rental income, grows the property-tax base, and deepens the city's innovation ecosystem as firms scale locally.

UC Berkeley's innovation and entrepreneurship ecosystem itself adds further economic value. Events hosted by Haas and other **campus programs attract visitors who spend on dining, transportation, and local services**, while engaging residents in the city's economy. These visitors frequently stay in hotels within Berkeley, and therefore contribute directly to the city's revenues by paying the 12 percent transit occupancy tax.

Interview data underline these ties: **in FY24–25, about 65% of Bakar Labs' and 30% of SkyDeck's procurement went to Berkeley-based vendors**. Regular spending at partners such as Noah's Bagels, Sliver, Safeway, ACC Printers, and Slippery Fish Catering shows how campus entrepreneurship turns innovation momentum into direct economic activity in the surrounding community.

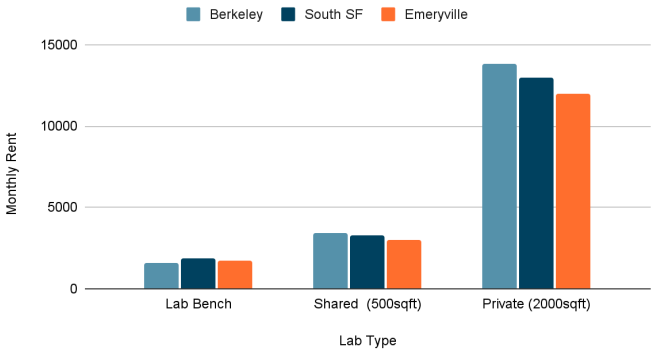
Opportunity #1: Space - Rental Rates & Accessibility



"Lots of people build really expensive bio space, and these types of spaces price all non-bio people out and we have to go elsewhere" - Founder, a non-biotech startup.

A common challenge for startups in Berkeley is the lack of affordable, appropriately sized laboratory and rental space. Many founders noted that the growth of biotech labs has increased competition for infrastructure, often at the expense of startups in other industries, such as hardware and robotics. These companies report difficulty finding space that meets specialized needs, as well as access to warehouse facilities and affordable truck services, limiting their ability to scale while remaining in Berkeley. Regardless, as Figure 4 demonstrates, Berkeley does not have a significant difference in rental rate compared to surrounding entrepreneurial cities. Using local posted prices for rental rates in each area, the average cost for Berkeley spaces was relatively lower for single benches and higher for large private spaces.[13]

Figure 4: Average Cost of Lab Space (Estimate)



Even biotech startups face constraints, as large facilities like Berkeley Commons can take 12–18 months to customize — longer than most startups' funding runways. Founders frequently cited South San Francisco, Emeryville, and Berkeley as long-term options, but trade-offs in cost, timelines, and convenience prevented consensus. While Berkeley's collaborative culture is highly valued, high costs, regulatory complexity, and limited space often push startups to locate elsewhere despite the city's strong innovation ecosystem.

Opportunity #2: Reducing intangible costs - Safety & Administration

The present uncertainty caused by public safety and street conditions, especially around Downtown Berkeley and BART stations, is a reason that led some founders to hesitate to start or renew leases.

[13] Findings aggregated from Bakar Labs, Bonneville Labs, Cushman Wakefield, & City of Berkeley

While the Bay Area produces many unicorns, startups' decisions to remain in Berkeley can be influenced by the surrounding urban environment. One program leader noted that employee perceptions of safety during commuting, office access, and late work hours may become a consideration over time. Despite Berkeley's strong resources and community, day-to-day city conditions can factor into long-term location decisions.

Streamlining access to regulatory and compliance information, such as through a Consolidated Regulatory Requirements Document, can improve the ease of doing business in Berkeley.

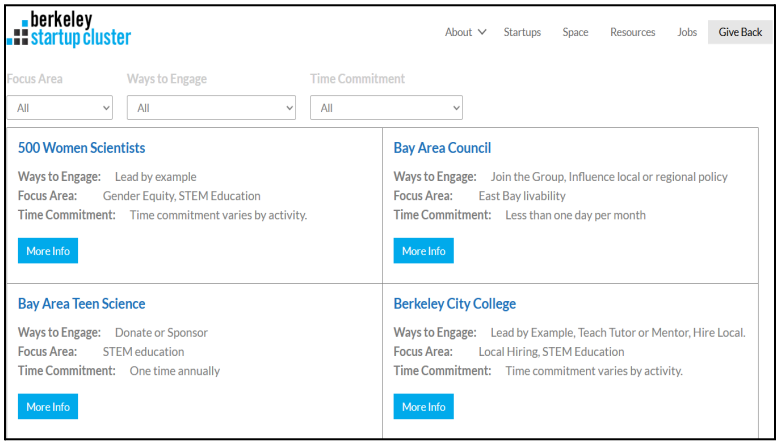
For early-stage startups, small fees and administrative requirements can create meaningful burdens. Founders reported spending significant time navigating Berkeley's licensing and permitting processes, and leaders from some programs noted that even modest fees or procedural complexity can influence location decisions. We believe these challenges relate more to administrative friction—such as timing and process complexity—than to the regulations themselves.

Startups also cited difficulty identifying and navigating compliance requirements, particularly during early formation. A consolidated digital portal outlining permitting, licensing, and regulatory processes could reduce uncertainty, especially for laboratory-based startups facing hazardous waste requirements.

Opportunity #3: Enabling Philanthropic Activities

When asked about contributing to local nonprofits or community initiatives, most startups reported limited past involvement but expressed strong interest, particularly in mentoring local students interested in entrepreneurship. A few founders cited volunteering with Berkeley Unified School District or UC Berkeley programs.

A resource to support this interest already exists through the Berkeley Startup Cluster's *Berkeley Ventures, Berkeley Values (BV2)* program, which provides a searchable list of local organizations for volunteering and donations. However, most founders interviewed were unaware of this resource.



Increasing awareness represents a low-cost opportunity. Coordination between the City of Berkeley and innovation ecosystem leaders could improve visibility through incubator

postings, brief mentions at campus I&E events, and targeted email reminders—such as on Giving Tuesday—encouraging startups to engage when they are ready.

6. CONCLUSIONS & CONSIDERATIONS

While it may seem intuitive that UC Berkeley is a main driver of innovation and entrepreneurship in the City of Berkeley, this report is the first of its kind to assess the impact of the campus's I&E and affiliated startups on the city. In our study, we identified the city-located startups that were affiliated with UC Berkeley and evaluated the ecosystem of campus I&E programs that support them from ideation to revenue generation. We then used surveys and subsequent analysis via IMPLAN to estimate the economic impact of startup activity on the City of Berkeley. Our results indicate that this **activity has created 516 jobs and driven slightly more than $200 million of added value to the city's economy**.

Moreover, conversations with startup founders, their employees, and leaders of campus I&E programs confirmed that UC Berkeley's innovation activity positively impacts the city in multiple ways:

- **Startups and I&E programs contribute directly to the local economy** by paying for the services of local restaurants, printing companies, and facility maintenance professionals.
- Existing startups, along with their employees and funding, are **drawn to the Berkeley area because of the entrepreneurial and R&D resources** that are available on campus.
- Founders of successful on-campus startups often **first look at office and lab space within the City of Berkeley** because of their familiarity with the area, supporting the city's commercial real estate market.
- The university **connects the city to a large network of skilled professionals and researchers**, enhancing the reputation of Berkeley as an innovative hub and drawing startups to Berkeley.
- Large conferences and **events held by I&E programs draw in visitors** who support local restaurants and contribute to the city's tax revenue through the 12 percent transit occupancy tax.

Considerations for the City of Berkeley

Our qualitative research unearthed several growth opportunities that we have condensed into four key recommendations for the City of Berkeley to consider incorporating into its innovation initiatives.

- **Improve perception of public safety:** While the City of Berkeley has made great strides in the last two years in reducing crime, in particular with commercial burglaries decreasing significantly for the second straight year, many founders we interviewed who left Berkeley cited safety concerns around downtown. Continuing to make substantive progress on crime while also publicizing this success could help to retain startups in the city.

- **Partner with UCB to spread awareness of philanthropic opportunities:** Many UC Berkeley affiliated startups expressed interest in giving back to the community but were unsure how to do so. Coordinating with campus I&E leaders to promote programs such as the Berkeley Startup Cluster's *Berkeley Ventures, Berkeley Values* program could substantially increase community involvement.

- **Streamline regulatory processes and information:** A common difficulty encountered by startups was finding relevant regulatory information to ensure they were compliant with City of Berkeley policies. Improved visibility and direct distribution of existing resources, such as the business license registration guide, would increase awareness of relevant regulations among startups. Additionally, creating a centralized online portal for permit and business license applications would further ease compliance, reduce administrative burden, and potentially improve startup retention and revenue for the city.

- **Increase accessible short-term lease office and lab space:** Actively seeking out and incentivizing commercial spaces that are smaller and easily customizable would directly address a challenge faced by many startups who struggle to find affordable lab and office space that meets their technical needs. Current options like the Berkeley Commons are too large and require too much time to customize for startups that are time and funding constrained.

Considerations for UC Berkeley

UC Berkeley should also encourage startups from all industries to become involved in the I&E ecosystem on campus, but in particular promote supportive efforts for biotech and software through campus resources and programs.

- **Invest in R&D:** Strong areas for the innovation ecosystem leaders to invest in include R&D, both through the university and in the surrounding areas focused on climate, biotech, medtech, and applied research technologies to expand the industrial base that the UC Berkeley-affiliated startups are nurturing in the city.

- **Invest in short-term lease rental space for startups organized by Berkeley professors, students, and network:** To foster entrepreneurship and a

commitment to startup culture, the university should emphasize room for innovation by providing necessary space for those who are starting ventures from Berkeley. This includes teams of students, faculty, professors, and even alumni.

Considerations for Future Studies

Finally, over the course of our research we also encountered some important questions that were outside the scope of this report but may be useful for future researchers to consider when examining innovation at UC Berkeley or the City of Berkeley:

- Why are some startups investing in Emeryville or Oakland based vendors instead of Berkeley vendors for services such as catering and housing?
- How do biotech versus non-biotech startups benefit from UC resources?
- Are there startups from a specific industry that are moving to Berkeley and/or staying permanently?
- What neighborhoods within Berkeley are most attractive to startups? Does this vary by industry?
- How do founders of the recent surge of AI startups view being based in Berkeley compared to other Bay Area cities, such as San Francisco?
- What percentage of federal and state taxes paid by startups reenters the local Berkeley economy in the forms of grants, subventions, and welfare distributions to Berkeley residents?
- When startups leave Berkeley, what are the most common places they relocate to, both in the Bay Area or elsewhere?

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Appendix A: Surveys

Startup Survey

These are the survey questions to collect information from UC Berkeley-affiliated startups.

1. Email address
2. What is your company's name?
3. In which of the following ways is your company connected to Berkeley, CA? *(Select all that apply.)*
    - I run my business out of a space on UC Berkeley's campus or another UCB-owned building.
    - I own or lease a Berkeley space (e.g., office, lab, workshop, etc.) off-campus.
    - I live in Berkeley and work from my home.
    - I have no physical space in Berkeley but use a Berkeley business address (PO Box, accelerator address, etc.).
    - Members of my company in leadership positions work in Berkeley, but I do not.
    - I studied or worked at UC Berkeley, but don't do business in Berkeley anymore.
    - Other:
4. If you have a work presence in Berkeley, what is/are your address(es)? *(Address, City, ZIP Code)*
5. How many Full-Time Equivalent (FTE) employees does your company currently employ in total?
   (Please exclude short-term contractors/interns unless they work with you on a long-term, ongoing basis and you consider them part of your core team.)
6. How many FTE work in Berkeley 2 or more days per week?
7. Approximately how much would it cost annually to hire your next five employees? *(Please estimate the fully loaded annual cost (salary + benefits) for the next five roles you expect to hire.)*
8. Approximately how much does your company pay annually to suppliers/vendors across all locations?
    - Less than $10k
    - Between $10k and $49k
    - Between $50k and $99k
    - Between $100k and $249k
    - Between $250k and $499k
    - Over $500k

9. Approximately how much funding has your company raised in the last fiscal year (inclusive of grants, equity funding, and debt financing)?
   (Note: All responses will be kept confidential.)
10. Approximately how much did your company spend on capital expenditure in the past fiscal year?
    (Please include equipment purchases, fit-out of office or lab space, facility upgrades, and large one-time investments in product development or infrastructure. Please do not include regular operating expenses such as salaries, rent, utilities, or routine software subscriptions.)
11. Has your business contributed to community or non-profit organizations in Berkeley (financial contributions, volunteer hours, mentorship programs, etc.)? If so, please describe.
12. On a scale of 0–10, with 0 being the least likely and 10 being the most likely, how likely are you to recommend Berkeley as a place to launch or grow a startup?
    (0–10 scale options provided.)
13. Would you be willing to speak with someone to share more about your company's Berkeley operations and local impacts? If so, please provide your preferred contact information for us to schedule a call.
14. Optional: Do you have any ideas or recommendations on how the City of Berkeley could be more accommodating to support the success of startups located within the city?

I&E Program Survey

1. Department/Unit Name
2. List the program(s) covered in this response
3. Primary contact name
4. Primary contact email address
5. Fiscal Year reported: set to FY24/25
6. Headcount (all) - Number (integer). Total count including part-time appointments (any %)
7. Personnel FTE- Number (decimal), sum of FTE across personnel.
8. Budget (Total, $), Include all fund types you directly manage.
9. Payroll (Total, $) - Gross wages paid by your unit.
10. Purchasing - local spending (Berkeley Vendors, $)
11. Purchasing - Total (all vendors, $) - *Total payments to all vendors*
12. If local vendor spend is unknown, estimate the % of total purchasing likely within Berkeley (Percentage)
13. Revenues - Grants ($), Sponsored research and grants.

14. Revenues - Gifts/Sponsorships ($), Philanthropy and corporate sponsorships
15. Revenues - Program Fees ($), Tuition/enrollment fees/event registration.
16. Revenues - IP/Licensing/Royalties ($) - Licensing revenue attributable to your unit.
17. Programs - workshops (#) - Distinct training sessions delivered.
18. Programs - Cohorts/accelerators (#) - Number of formal cohort-style programs run.
19. Programs - Participant count (#) - Total participants served across trainings/programs
20. Startups launched in FY - New entities formed by participants in the year.
21. Internships/Jobs placed in Berkeley (#)
22. What share of your staff/students likely live in Berkeley (%)?
23. Notable Berkeley-based vendors or partners (list)
24. May we follow up to clarify responses? Yes/No

Appendix B: IMPLAN Modeling Framework

B.1 Modeling Objective

IMPLAN was used to estimate the economic impacts generated by Berkeley-affiliated startups through employment, supplier spending, and household consumption effects. The analysis focuses on operational activity and reports impacts as the combined effect of direct, indirect, and induced channels.

Because survey responses were obtained from 34 firms out of an identified ecosystem of 182 startups, modeled results represent verified observed impacts that later serve as the empirical foundation for ecosystem extrapolation.

B.2 Data Preparation

Survey responses were cleaned and standardized prior to modeling. The following variables were extracted for IMPLAN input:

- **Berkeley FTE:** Employees working two or more days per week in Berkeley, modeled as full-time equivalent employment.
- **Vendor spending:** Ongoing operational expenditures, modeled as Industry Output.
- **IMPLAN sector assignment:** Each firm was manually assigned to an IMPLAN sector based on primary business activity.

- **Tax compliance status:** Presence or absence of a Berkeley business license.

Firms with no verified local operating presence were excluded. Duplicate survey entries were consolidated. An internal sector crosswalk was maintained to ensure transparency and replicability, recognizing that many startups span hybrid software, research, and professional service activities.

Survey Sample Characteristics

The 22 modeled firms exhibit substantial heterogeneity typical of early-stage startup ecosystems:

Table B.2-1: Descriptive Statistics of Survey Sample (22 firms)

Metric	Min	Max	Mean	Std. Dev.	CV	Interpretation
Berkeley FTE	0	33	3.1	6.8	2.19	High variance indicates presence of anchor firms alongside small startups
Total FTE	0	100	9.9	20.0	2.02	Suggests heavy-tailed firm size distribution
Vendor Spending	<$10K	>$500K	$181,406	$226,756	1.25	Right-skewed distribution typical of early-stage ecosystems
Capital Expenditure	$0	$5M	$224,359	$910,678	4.06	Extremely high variance reflects episodic, firm-specific investment patterns

Coefficient of Variation (CV) = Standard Deviation / Mean

Key Observations:

1. High coefficients of variation (CV > 1.0 for all metrics) confirm heavy-tailed distributions and validate the segmented extrapolation approach (Appendix C) over simple linear scaling.
2. FTE ratio (Total FTE / Berkeley FTE ≈ 3.2) suggests firms employ approximately 2.2 additional workers outside Berkeley for every Berkeley-based employee, consistent with distributed workforce patterns in tech and R&D sectors.
3. CapEx variability (CV = 4.06) validates the decision to exclude capital investment from extrapolation framework, as the extreme variance makes averaging inappropriate for ecosystem-wide estimates.

B.3 Firm Inclusion Rules

Each firm was classified into one of three modeling categories:

- **Include (Employment + Output):** Firms reporting Berkeley-based FTE and vendor spending (17 firms). Modeled using both Industry Employment and Industry Output events.
- **Output-only:** Firms without Berkeley FTE but with verified local vendor spending or lab/office activity (5 firms). Modeled using Industry Output only.
- **Exclude:** Firms reporting no local employment, no identifiable local operational or capital activity, or inconsistent survey responses (12 firms). Not modeled.

This classification avoids double counting and ensures that impacts reflect verified Berkeley-based economic activity.

B.4 IMPLAN Event Structure

Two operational IMPLAN event types were applied:

- **Industry Employment:** Berkeley-based workers.
- **Industry Output:** Ongoing vendor spending.

Vendor spending followed IMPLAN's default regional purchasing assumptions. Because early-stage startups often procure specialized services outside the immediate region, indirect effects may be modestly overstated. This limitation is addressed in the interpretation section and motivates future sensitivity analysis.

B.5 Geography and Commuting

IMPLAN modeling was conducted at the **Alameda County** geographic level due to dataset resolution constraints. Although employment inputs were localized to Berkeley-based workers, induced household spending and supplier linkages were allocated across the broader Alameda County regional economy.

Accordingly, induced effects should be interpreted as regional spillovers generated by Berkeley-based startup activity rather than spending confined strictly within city boundaries.

B.6 Scenario Structure for Observed Firms

Two operational scenarios were constructed:

- **Scenario A - Compliant Firms:** Only firms with verified Berkeley business licenses.
- **Scenario B - All Surveyed Firms:** All modeled firms regardless of tax compliance.

The difference between these scenarios provides an indicative tax exposure gap. However, IMPLAN tax results reflect sectoral averages and should not be interpreted as literal unpaid obligations.

The distinction between Scenario A (Compliant) and Scenario B (All Firms) is relevant only for fiscal impact interpretation (see Appendix E). All extrapolation ranges in the main report (Table 1) are based exclusively on Scenario B operational impacts and do not differentiate by compliance status.

IMPLAN tax estimates represent modeled tax incidence based on sectoral averages rather than verified municipal collections. For interpretive context, future work should compare IMPLAN sub-county general tax estimates with reported City of Berkeley business license revenues and proportional sales tax collections to better align modeled incidence with observable revenue streams.

B.7 Capital Expenditure Modeling

Reported capital investments were modeled using IMPLAN's Institutional Spending Pattern for Capital Formation.

This pattern distributes investment across construction, equipment manufacturing, installation services, and professional services using standard investment profiles. Regional purchasing coefficients reflect that construction and installation are primarily local, while specialized equipment is often sourced nationally.

Table 2: Regional Economic Footprint of Capital Investment[14] - Alameda County

Metric	CapEx survey impact*	Operational Survey impact
Employment	20	220
Labor Income (Millions)	$2.26	$35.69
Value Added (Millions)	$3.38	$63.22
Output (Millions)	$5.87	$93.67

[14] *Capital investment impacts should be interpreted as one-time investment-related effects rather than as recurring annual contributions to the regional economy.*

Table 3: *Survey Economic Impacts* of Capital Investment[15] - City of Berkeley (Direct impacts)

Metric	CapEx survey impact*	Operational Survey impact
Jobs in Berkeley	13	107
Annual Payroll (Millions)	$1.61	$23.20
Value Added (Millions)*	$4.10	$61.84

B.8 Interpretation of Observed Results

IMPLAN results demonstrate that even a subset of Berkeley-affiliated startups generates meaningful employment, income, output, and fiscal effects. However, results likely understate the ecosystem's full footprint due to:

- Non-responding firms,
- Conservative local purchasing assumptions,
- Separate treatment of capital investment,
- Geographic leakage of household spending.

Conversely, some indirect effects may be modestly overstated where startups rely on specialized non-local vendors.

The observed results therefore serve as a verified baseline rather than a full accounting of Berkeley's innovation economy.

B.9 IMPLAN Results

Scenario A: COMPLIANTS

Taxation Operational impact

Impact	Sub County General	Sub County Special Districts	County	State	Federal	Total
1 - Direct	$111,558	$107,066	$65,208	$776,954	$2,366,798	$3,427,584
2 - Indirect	$44,279	$42,398	$25,821	$224,885	$706,580	$1,043,963
3 - Induced	$78,778	$74,930	$45,627	$214,079	$423,580	$836,993
Total	$234,614	$224,395	$136,656	$1,215,917	$3,496,958	$5,308,540

[15]

Scenario B: ALL FIRMS (Survey)

Operational economic impact

Impact	Employment	Labor Income	Value Added	Output
1 - Direct	107	$23,195,196	$42,040,713	$61,836,344
2 - Indirect	59	$8,074,557	$12,756,443	$19,120,691
3 - Induced	54	$4,418,096	$8,422,860	$12,417,396
Total	220	$35,687,849	$63,220,016	$93,374,431

Taxation: Operational impact

Impact	Sub County General	Sub County Special Districts	County	State	Federal	Total
1 - Direct	$227,928	$218,956	$133,357	$1,575,137	$5,094,475	$7,249,853
2 - Indirect	$105,232	$100,752	$61,359	$518,879	$1,642,864	$2,429,085
3 - Induced	$176,209	$167,603	$102,057	$478,835	$947,411	$1,872,115
Total	$509,370	$487,311	$296,773	$2,572,851	$7,684,749	$11,551,054

Economic impact: CAPEX

Impact	Employment	Labor Income	Value Added	Output
1 - Direct	13	$1,609,254	$2,242,258	$4,102,734
2 - Indirect	4	$379,842	$633,094	$1,014,111
3 - Induced	3	$266,760	$508,562	$749,741
Total	20	$2,255,856	$3,383,914	$5,866,586

Appendix C: Extrapolation Framework

C.1 Purpose

Because only a subset of Berkeley-affiliated startups provided usable survey data, an extrapolation framework was developed to approximate ecosystem-wide economic impacts. The objective is not to produce a census estimate, but to generate a defensible, evidence-based impact range grounded in observed data and transparent assumptions.

C.2 Conceptual Structure

Startup ecosystems do not follow linear or normal distributions. Empirical evidence consistently shows that innovation systems exhibit heavy-tailed behavior, where a small number of firms account for a disproportionate share of employment, output, and value creation.

To reflect this structure, the extrapolation framework adopts a two-segment model:

- **Anchor firms:** large, high-employment startups treated as directly observed.
- **Long-tail firms:** the broader population of smaller startups representing the typical ecosystem participant.

Accordingly, the methodology:

1. Treats anchor firms as directly observed and not subject to scaling.
2. Models the remaining firms as a long tail.
3. Scales only the long tail to the full ecosystem.
4. Recombines anchors and extrapolated tail to estimate total ecosystem impacts.

This structure preserves the empirical distribution of firm sizes and avoids distortions caused by linear averaging.

C.3 Relevant Populations

Category	Count	Description
Identified startup ecosystem	182	All verified startups linked to Berkeley
Survey respondents	34	Firms providing usable responses
Firms with verifiable local activity (modeled in IMPLAN)	22	Firms with verifiable Berkeley activity
Firms excluded from modeling	12	No confirmed Berkeley footprint

Firms excluded from IMPLAN modeling because of insufficient local activity data remain included in the ecosystem population count under the assumption that they represent legitimate but currently low-activity or early-stage ventures. Excluding these firms from the extrapolation base would risk understating the true ecosystem size.

Within the 22 modeled firms:

- **Anchor firms:** 2 firms (≥20 Berkeley FTE)

- **Observed tail firms:** 20 firms

Thus, the ecosystem long-tail population equals:

181=182−2

C.4 IMPLAN Inputs for Extrapolation

To isolate long-tail behavior, IMPLAN was re-run excluding the two anchor firms. The resulting tail-only operational impacts were:

Metric	Tail-only Impact City of Berkeley (Direct impacts)	Tail-only Impact Alameda County
Employment	51	104
Labor Income	$11.06	$16,980,329
Value Added	$20.07	$30,110,309
Output		$44,799,580
Taxes Sub county General Direct		$106,084

These values constitute the empirical foundation for extrapolation.

C.5 Scenario-Based Extrapolation Framework

Because startup ecosystems exhibit structural uncertainty and heavy-tailed dynamics, a scenario-based framework was adopted. Rather than producing a single estimate, two complementary scenarios define an evidence-based impact range.

Both scenarios rely on identical IMPLAN results for observed firms and differ only in how the unobserved portion of the ecosystem is treated.

C.6 Scenario 1: Conservative Lower Bound

Scenario 1 assumes that the observed long-tail firms are fully representative of the broader ecosystem and applies linear scaling without additional power-law correction.

In this scenario:

- Anchor firms are treated as directly observed and are not scaled.
- Only the long-tail sample is extrapolated.
- No additional contribution from unobserved high-performing firms is assumed.

Tail Scaling Factor: Tail Factor=181/20=9.05

This factor is applied exclusively to long-tail operational impacts.

Scenario 1 results: Lower extrapolation.

CITY OF BERKELEY IMPACTS (Direct) - Lower extrapolation					
Metric	**Tail (IMPLAN)**	**Tail factor**	**Ecosystem tail - Direct**	**Observed anchors**	**Total ecosystem**
Jobs in Berkeley	51	9.05	460	56	**516**
Annual Payroll (Millions)	$11.06	9.05	$100.13	$12.13	**$112.27**
Value Added (Millions)	$20.07	9.05	$181.59	$21.98	**$203.57**

REGIONAL ECONOMIC FOOTPRINT (Alameda County) - Lower extrapolation					
Metric	**Tail (IMPLAN)**	**Tail factor**	**Ecosystem tail**	**Observed anchors**	**Total ecosystem**
Employment	104	9.05	945	116	**1061**
Labor Income (Millions)	$16.98	9.05	$153.67	$18.71	**$172.38**
Value Added (Millions)	$30.11	9.05	$272.50	$33.11	**$305.61**
Output (Millions)	$44.80	9.05	$405.44	$48.57	**$454.01**
Sub county general direct	$0.11	9.05	$0.96	$0.12	**$1.08**
Sub county general total	$0.24	9.05	$2.19	$0.27	**$2.45**

Scenario 1 therefore represents a conservative lower bound of ecosystem impact.

C.7 Scenario 2 — Realistic Power-Law Adjustment

Rationale

Scenario 2 accounts for the empirical reality that startup ecosystems follow highly skewed size distributions. Under such distributions, high-impact firms are systematically underrepresented in limited survey samples.

Scenario 2 augments Scenario 1 by statistically estimating the expected contribution of unobserved high-impact firms (“missing top outcomes”) using Monte Carlo simulation.

Total ecosystem impact is therefore:

$$Scenario\ B = Extrapolated\ Tail + Observed\ Anchors + MissingTopAddOn$$

C.8 Statistical Estimation of Missing Top Firms

Dataset Preparation

Firm-level Berkeley employment was extracted for all startups with positive FTE. Firms with zero FTE were excluded from the statistical fit.

Lognormal Transformation

Employment values were transformed: LN_FTE= ln(FTE_inBerkeley)

Distribution parameters were estimated:

- μ=mean(LN_FTE)
- σ=std_dev(LN_FTE)

Given the limited number of observations, formal goodness-of-fit tests lack sufficient power to reliably validate the distributional assumption. Therefore, the lognormal specification is adopted based on well-established empirical evidence in the literature on firm size and startup growth patterns (See Appendix B.2).

Monte Carlo Simulation

Using the fitted lognormal distribution, Monte Carlo simulation generated synthetic firm employment values for a population of 182 startups.

Each iteration:

- Generated 182 firms,
- Ranked firms by employment,
- Recorded anchor-scale firms (≥20 FTE),
- Recorded top 3-10 firm sizes.

A total of 300 iterations were performed.

C.9 Sectoral Allocation of Missing-Top Firms

Observed high-employment firms are concentrated in:

- Software Publishing (IMPLAN 410)
- Scientific R&D Services (IMPLAN 446)

These sectors represent the upper tail of Berkeley's startup ecosystem and serve as the basis for allocating missing-top employment. The two anchors of the sample are from each one of the industries mentioned above.

Sensitivity Analysis and Scenario Averaging
Because the "true" sectoral composition of unobserved high-performers is unknown, a scenario-based averaging approach was adopted to avoid arbitrary allocation decisions.

The Monte Carlo simulation estimated 127 missing-top jobs (mean) that were not captured in the survey. Rather than assigning these jobs to a single sectoral allocation, three plausible scenarios were modeled separately in IMPLAN and then averaged.

Each scenario was modeled independently in IMPLAN using Industry Employment events. The resulting economic impacts (employment, labor income, value added, output) were then averaged across the three scenarios to produce the final missing-top contribution used in Scenario 2.

C.10 Integration with IMPLAN
Missing-top employment was modeled as Industry Employment events in IMPLAN across three sectoral allocation scenarios.

Each scenario produced a full set of economic impacts (direct, indirect, induced effects). Integration Process:

1. Run IMPLAN for each scenario:

- Scenario 1: 77 FTE in IMPLAN 410 + 50 FTE in IMPLAN 446.
- Scenario 2: 50 FTE in IMPLAN 410 + 77 FTE in IMPLAN 446.
- Scenario 3: 64 FTE in IMPLAN 410 + 63 FTE in IMPLAN 446 2.

2. Extract total impacts (direct + indirect + induced) for each scenario

3. Calculate average across the three scenarios for:

- Total Employment
- Labor Income
- Value Added
- Output

C.11 Scenario 2 Results: Higher extrapolation

CITY OF BERKELEY IMPACTS (Direct) - Higher extrapolation						
Metric	**Tail (IMPLAN)**	**Tail factor**	**Ecosystem tail**	**Observed anchors**	**Missing-top add-on**	**Total ecosystem**
Jobs in Berkeley	51	9.05	460	56	127	643
Annual Payroll (Millions)	$11.06	9.05	$100.13	$12.13	$26.19	$138.46
Value Added (Millions)	$20.07	9.05	$181.59	$21.98	$45.98	$249.55

REGIONAL ECONOMIC FOOTPRINT (Alameda County) - Higher extrapolation						
Metric	**Tail (IMPLAN)**	**Tail factor**	**Ecosystem tail**	**Observed anchors**	**Missing-top add-on**	**Total ecosystem**
Employment	104	9.05	945	116	258	1319
Labor Income (Millions)	$16.98	9.05	$153.67	$18.71	$40.51	$212.89
Value Added (Millions)	$30.11	9.05	$272.50	$33.11	$70.09	$375.70
Output (Millions)	$44.80	9.05	$405.44	$48.57	$104.30	$558.31

Scenario 2 represents the central, most realistic estimate.

C.12 Interpretation and Limits

Both scenarios represent structured approximations rather than a census. Remaining uncertainties include:

- Non-response bias,
- Firm churn,
- Verification limits,
- Non-extrapolated capital investment,
- Residual power-law variation.

Appendix D: Key Definitions

The economic impact model outlined below uses the terms set by IMPLAN, and in order to correctly interpret the outputs, it is necessary to understand the exact meaning and implications of the terms Value Added and Output. The definitions for both of these terms as described by IMPLAN are outlined below[16].

D.1 Value Added

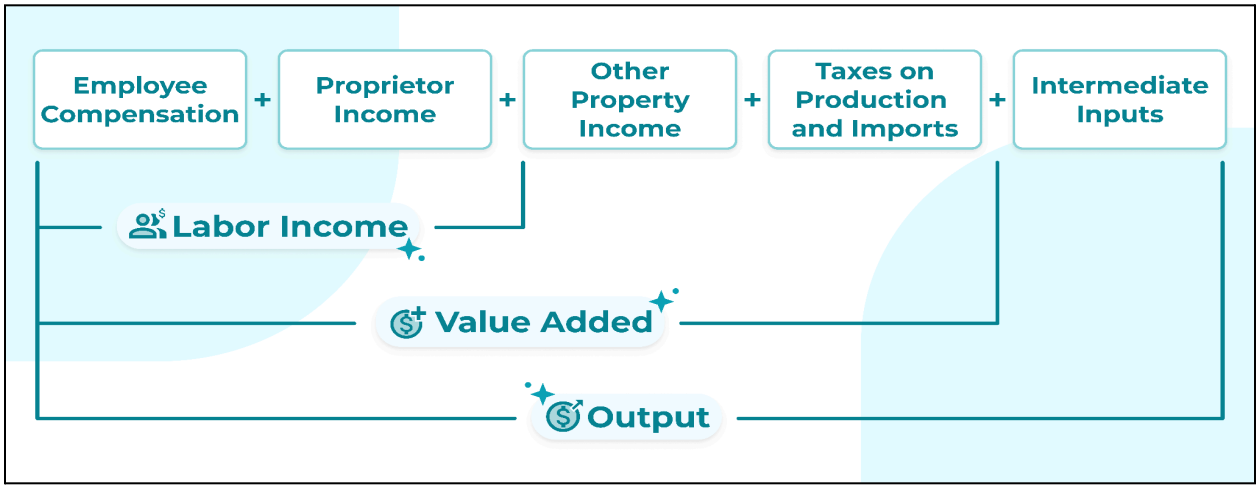
The total market value of all final goods and services produced within a region in a given year. Value Added is also a measure of an Industry's production, but with the costs of Intermediate Inputs removed. Value Added represents the incomes that are generated by production. Value Added can be measured in two different ways, both of which will produce the same results. It can either be measured by subtracting the cost of Intermediate Inputs from an Industry's total Output or by summing each Industry's payments to the components of Value Added: Labor Income (LI), Other Property Income (OPI), and (TOPI). In effect, Value Added represents the contribution of the Industry to the economy.

D.2 Output

A measure of the total value of all goods produced. Value Added is a subset of Output and is a useful measure of wealth created by an economy. An Industry buys goods and services from other Industries and remanufactures those goods and services to create a product of greater value (Output) than the sum of the goods that goes into its product (Intermediate Inputs). That increase in value is the value that the producer adds to the inputs as a result of the production process. This added value is then used to pay labor and taxes with hopefully some remainder for profit.

[16] These definitions and graphic are drawn directly from IMPLAN and can be found at https://support.implan.com/hc/en-us/articles/360025171053-Output-Value-Added-and-Double-Counting

See the figure below for a visual representation of what is included in each term.



Appendix E: Tax Impact Interpretation and Limitations

E.1 Nature of IMPLAN Tax Estimates

IMPLAN tax outputs represent modeled **tax incidence**, not verified tax collections. Tax estimates are calculated using sector-average effective tax rates applied to modeled labor income, production, and consumption flows. As a result, IMPLAN tax results reflect the distribution of tax burdens across levels of government rather than actual payments made by individual firms.

Tax categories reported by IMPLAN include:

- Sub-County General
- Sub-County Special Districts
- County
- State
- Federal

These categories combine multiple tax instruments such as sales and excise taxes, property-related taxes, income taxes, payroll-related taxes, and selected business-related fees.

Accordingly, IMPLAN tax results should not be interpreted as direct fiscal revenues collected by any specific jurisdiction.

E.2 City of Berkeley Interpretation

This modeling includes all 22 firms with verifiable Berkeley economic activity regardless of business license status. This approach captures the fiscal incidence associated with actual economic activity, not verified tax compliance.

The tax estimates below reflect **modeled tax incidence** based on economic activity (employment, payroll, spending). They should be interpreted as:

- Potential fiscal contribution if all firms were compliant.
- Economic activity scale for policy planning purposes.
- NOT actual verified tax payments to the City of Berkeley.

Because only 2 of 22 modeled firms had verified business licenses, actual municipal tax collections are likely substantially lower than modeled estimates.

Methodological Choice: All Economic Activity vs. Verified Compliance

This appendix reports tax incidence based on Scenario B (All Firms) to capture the fiscal scale associated with observed economic activity. However, it is critical to distinguish:

- Modeled tax incidence: $228K direct (all 22 firms with economic activity).
- Verified compliant contribution: ~$112K direct (2 firms with licenses).
- Non-verified activity: ~$116K direct (20 firms without verified licenses).

The $116K difference should NOT be interpreted as:

- Unpaid taxes or tax evasion.
- Recoverable municipal revenue.
- Evidence of intentional non-compliance.

It may reflect:

- Early-stage firms not yet meeting licensing thresholds.
- Administrative delays in license processing.
- Firms operating remotely with only UC Berkeley affiliation.
- Legitimate exemptions from certain license requirements.
- Actual non-compliance (unknown proportion)

For fiscal planning and budgeting, the verified compliant figure ($112K) represents the most conservative estimate. For economic development analysis, the total economic activity figure ($228K) provides a more complete picture of startup ecosystem fiscal footprint.

Geographic Attribution of Tax Incidence: Because IMPLAN models economic impacts at the Alameda County level, indirect and induced effects (and their associated tax incidence) are distributed across multiple municipalities within the county. Only the direct component of Sub-County General taxes can be reasonably attributed to Berkeley, as direct impacts reflect economic activity physically occurring within city limits.

Impact	Sub County General (Scenario B, All firms)	Geographic Attribution
Direct	$227,928	Primarily City of Berkeley (Only ~$112K from verified compliant firms)
Indirect	$105,232	Regional suppliers (distributed across county)
Induced	$176,209	Regional household spending (distributed across county)
Total	$509,369	Alameda County (multiple jurisdictions)

The direct Sub-County General tax incidence of approximately $228,000 represents the most defensible proxy for Berkeley-attributable municipal tax contribution from surveyed startups. This figure captures taxes directly associated with:

- Berkeley-based employment (payroll-related taxes).
- Berkeley business operations (business license fees, property tax on occupied space).
- In-city purchases and transactions.

The indirect ($105K) and induced ($176K) components, totaling $281K, represent tax incidence distributed across other Alameda County municipalities where supplier activities and household spending occur.

Only 2 of the 22 modeled firms had verified Berkeley business licenses. Based on Scenario A (compliant firms), these 2 firms account for approximately $112,000 of the $228,000 direct tax incidence, suggesting that verified compliant contribution represents roughly 49% of total modeled direct taxes.

This represents economic activity from 22 modeled firms. Extrapolating to the full ecosystem using the same methodology applied to employment and output:

Estimated Berkeley-Attributable Municipal Tax Incidence:

Scenario	Direct Sub-County General
Survey	$227,928
Lower extrap.	$1,081,000
Higher extrap.	$1,357,000

Even this refined estimate reflects modeled sectoral averages, not actual Berkeley tax collections. IMPLAN cannot distinguish between:

- Business license fees vs. other local taxes.
- Sales tax (city share) vs. property tax vs. utility users tax.
- Compliant vs. non-compliant firms (only 2 of 22 had verified licenses).

Accordingly, these values should be interpreted as **order-of-magnitude indicators** of fiscal scale rather than precise revenue forecasts.

The following table shows the direct tax incidence jurisdiction scenario B (All Firms) - Survey Sample:

Tax Category	Direct	Indirect	Induced	Total
Sub County General	$227,928	$105,232	$176,209	$509,370
Sub County Special Districts	$218,956	$100,752	$167,603	$487,311
County	$133,357	$61,359	$102,057	$296,773
State	$1,575,137	$518,879	$478,835	$2,572,851
Federal	$5,094,475	$1,642,864	$947,411	$7,684,749
Total	$7,249,853	$2,429,085	$1,872,115	$11,551,054

E.3 Compliance-Based Modeling

Tax impacts were modeled only for firms with verified Berkeley business license compliance. Because only two of the twenty-two modeled firms met this criterion, tax estimates reflect compliant firms only and should be interpreted as conservative lower-bound indicators of fiscal contribution.

This approach avoids overstating fiscal impacts while preserving methodological transparency.

E.4 Survey-Based Tax Results

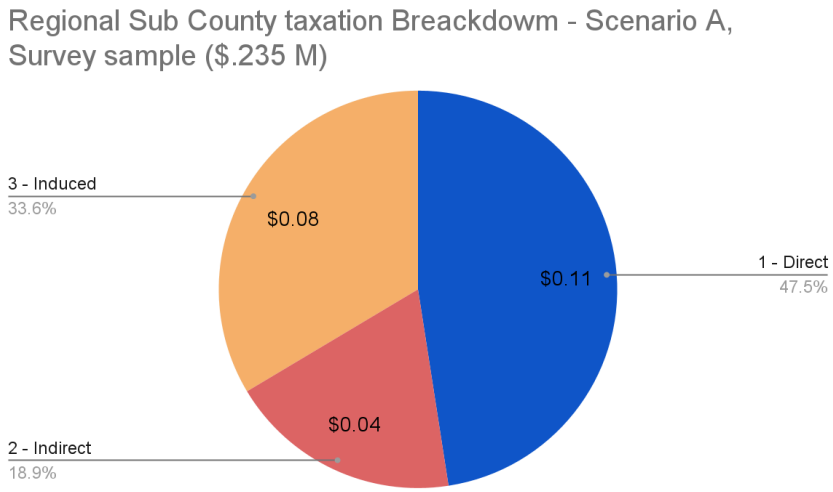
This section presents tax incidence estimates at two levels: (1) verified compliant firms only, and (2) all observed economic activity.

E.4.1 Verified Compliant Firms (Conservative Estimate)

Based on 2 firms with verified Berkeley business licenses (Scenario A):

- Total tax incidence (all jurisdictions): $5.31 million annually
- Sub-County General (municipal proxy): $234,614
- Direct (Berkeley-attributable): $111,558

This represents the most conservative estimate of verified fiscal contribution from surveyed startups.

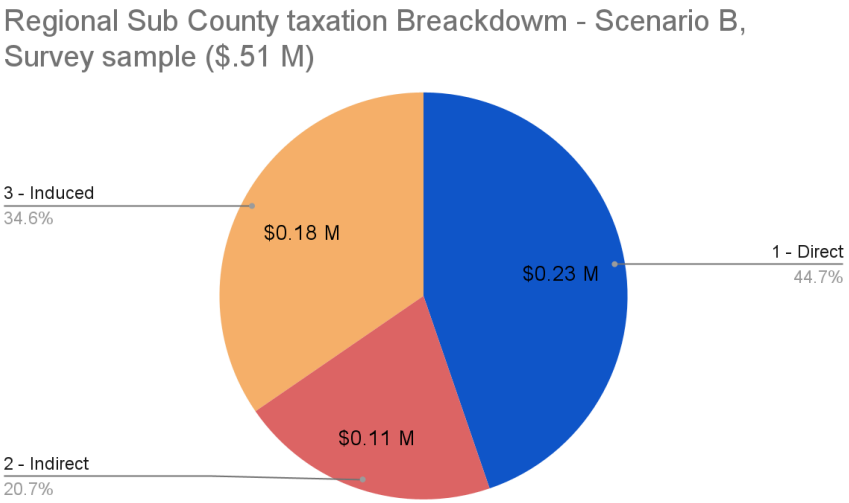


E.4.2 All Economic Activity (Full Sample)

Based on all 22 firms with verifiable Berkeley economic activity (Scenario B):

- Total tax incidence (all jurisdictions): $11.55 million annually
- Sub-County General (municipal proxy): $509,370
- Direct (Berkeley-attributable): $227,928

This represents the fiscal incidence associated with all observed economic activity, regardless of license compliance status.



E.4.3 Compliance Gap Analysis

Tax exposure gap (Berkeley specific)

Item	Scenario A (Compliant firms)	Scenario B (All firms)	GAP
Sub County General	$111,558	$227,928	$116,370

The $116,370 difference reflects economic activity by firms without verified business licenses. This should be interpreted as:

- **Order of magnitude** of fiscal incidence from non-verified activity.
- **NOT recoverable revenue** firms may have legitimate reasons for lacking licenses.
- **Planning metric** for understanding the relationship between startup activity and municipal fiscal capacity.

E.4.4 Extrapolated Berkeley Municipal Tax Estimates

Applying the same extrapolation methodology used for employment and output to the direct Sub-County General tax incidence:

Taxation Extrapolation		
Metric	**Lower Extrap.**	**Hieguer Extrap.**
Sub county general direct	**$1.08**	**$1.36**
Sub county general total	**$2.45**	**$3.06**

Berkeley-affiliated startups likely generate $1.08 - $1,36 million in municipal tax incidence annually under full ecosystem extrapolation. This represents an order-of-magnitude estimate based on IMPLAN sectoral averages and should be validated against actual city revenue data where possible.

E.5 Interpretation Guidance

IMPLAN tax estimates should be interpreted as:

- Indicators of fiscal scale associated with startup economic activity,
- Not as direct measures of municipal tax collections,
- Not as evidence of tax non-compliance or underpayment by specific firms.

For Berkeley specifically:

- Use: Direct Sub-County General.
- Caution: Total Sub-County General (~$509K observed) - includes regional spillovers.
- Avoid: Total tax incidence (~$11.6M) - primarily state/federal.

Future work should compare IMPLAN modeled tax incidence with:

- City of Berkeley business license revenue (actual collections).
- Sales tax collections (city share) - Proportional allocation of county and state tax streams.
- Property tax receipts from startup-occupied commercial space.

Such triangulation would improve fiscal attribution accuracy and allow validation of IMPLAN sectoral assumptions against Berkeley-specific data.

Appendix F: Limitations and Needs

This analysis provides a robust evidence-based estimate of startup economic impacts, but several limitations should be acknowledged:

Methodological Limitations

1. Geographic Resolution: IMPLAN's smallest available geography is Alameda County, preventing direct modeling of multiplier effects at the Berkeley city level. As a result:
- City-level results are limited to direct activity only.
- Spillover effects (indirect/induced) are modeled regionally.
- True city-retained economic activity is likely between direct-only and total regional estimates.
2. Tax Incidence Modeling: IMPLAN tax results represent modeled sectoral averages and cannot be interpreted as actual City of Berkeley revenue collections. Firm-level tax compliance varies, and only 2 of 22 modeled firms had verified business licenses at the time of analysis. Tax estimates (Appendix E) should be viewed as indicative only.

Data Limitations

1. Survey Coverage: Only 34 of 182 identified firms (19%) provided usable survey responses, limiting precision. While extrapolation scenarios address this gap, residual uncertainty remains regarding:
- Non-response bias (are respondents representative?).
- Accuracy of self-reported data.
- Ecosystem boundary definition (who qualifies as "Berkeley-affiliated"?)

2. Firm Churn: Startup ecosystems exhibit high entry/exit rates. This analysis provides a point-in-time snapshot but cannot account for:
- Startups founded after data collection.
- Firms that ceased operations during the study period.
- Rapid growth/decline between survey and analysis.

Appendix G: Methodology Learnings

Methodology Learnings

On the data collection side of the project, attending events is vital to having the necessary components of *who startups are.* This is where the heart of the project exists— in the conversations with founders on their experience and hopes for the city of Berkeley. Using a multiple-person approach of interviewing startups to gather information and clarify ambiguities throughout each week is a necessary implementation

that our team discovered later in the process. Targeting different themes and directions of interviews is important. When analyzing responses, one should consider numerical results alongside personal accounts, integrating empirical evidence with interpretive understanding to guarantee a well-rounded perspective into the lives of startups.

Potential Solutions:

Through the research process, the research team encountered technological, timeline, and storytelling challenges. By effectively identifying holes in the study, potential future iterations may use the following suggestions for more efficient, thorough, and accurate results.

a. **Build a digital portal**

   Through our own process, we determined that building a digital portal to integrate entrepreneurial resources within the city's governmental approvals would be beneficial, as well as a digital identity to track projects throughout their process of moving to the city. Additionally, observing the amount of federal taxes paid by a company that reenters Berkeley's economy in the form of federal grants is a promising project that could be completed in more time. Berkeley-based vendor and operating expenditures constitute the primary operational inputs for IMPLAN employment and output events. This project could be led by researchers collecting this data.

b. **Survey Design and Distribution**

   Through the survey process, we found that startups were less likely to respond when the survey appeared lengthy. To maximize response rates, the survey was limited to 10 clearly worded questions to keep it approachable. On the distribution side, framing the research as aiming to help the I&E ecosystem for startups can improve response quality. Cold outreach yielded limited results, while responses were strongest when departments leveraged existing relationships to invite participation.

c. **Interviewing startups**

   On the data collection side, attending events proved essential to understanding who the startups are and how they operate. The core of the project lies in direct conversations with founders. This is where the heart of the project exists— in the conversations with founders on their experience and hopes for the city of Berkeley. Using a multiple-person approach of interviewing startups to gather information and clarify ambiguities throughout each week is a necessary implementation that our team discovered later in the process. Targeting different

themes and directions of interviews is important. When analyzing responses, one should consider numerical results alongside personal accounts, integrating empirical evidence with interpretive understanding to guarantee a well-rounded perspective into the lives of startups.

d. Alternative approaches to IMPLAN

We discovered that using the economic software IMPLAN was limiting for our research scope, as it can examine at the county level but not the city level. This was extremely limiting, and urged us to consider whether there is another way to measure independent economic impact of cities rather than the analysis of counties for future studies? Evaluating the economic impact of the startups on the city of Berkeley rather than Alameda County would be more accurate. Additionally, because IMPLAN tax results represented modeled tax incidence rather than municipal collections, future studies should seek to collect firm-level data on business license payments, local sales tax generation, and other Berkeley-specific tax streams for accuracy. Increased survey coverage and repeating data collection would allow for more precise results given more time and data. Lastly, given the heavy-tailed nature of startup ecosystems, we should separate anchor firms from the long-tail population, extrapolating only the tail while preserving observed anchor impacts. A Monte Carlo simulation based on firm-level FTE distributions provides a defensible approach to estimating the presence and size of unobserved high-performing firms.

APPENDIX H: UC Berkeley-Affliated Startups in Berkeley, CA STARTUPS, 2025

SOFTWARE

1. Agentero
2. Autoderm
3. Blacktop Hoops
4. Blind
5. Boxlet Storage
6. Bream
7. CarbonSustain
8. Code Blue AI
9. Cycle
10. Copper
11. Cycle
12. Datakin
13. De Mobo
14. Deeli AI
15. Diagnoss
16. Diffit
17. Elevate Security
18. Elodin
19. EvoShare
20. Greaten AI
21. Intertrust Technologies
22. Jacobi Robotics
23. Kintsugi
24. Ladon
25. MachineVantage
26. Melanin Meanings
27. Meso (Other Financial Services)
28. New Avenue
29. Reach Best
30. Scribble AI
31. SigIQ.ai
32. Simple But Needed
33. SparcIt
34. WideSense
35. Xronos
36. AtlasNova
37. Clouddir
38. Flexify.AI
39. Guepard
40. Numix
41. OZARU
42. Pickaxe
43. VoiceBit Inc
44. Wing Assistant
45. Rumi
46. Papaya
47. Aerosol Dynamics Inc.
48. AIMATX
49. Bepress
50. Imagize LLC
51. SiFive

BIOTECH

52. 712 North
53. Bikanta
54. BioGlyph
55. BluumBio
56. Calcott Consulting
57. Correlia Biosystems
58. Elara Biosciences, LLC
59. Foresight
60. Foundry31
61. Gisens Biotech
62. Glyphic Bio
63. HexemBio
64. HypO2Regen Therapeutics
65. KarnaTeq
66. Keylika
67. Lumiphore
68. Magnetic Tides
69. Million Marker
70. Protocols.io
71. Renegade Bio
72. Rhea's Factory
73. Ribogenics Technologies
74. Valitor Bio
75. CatenaBio
76. Maxivio Inc
77. MoAb Therapeutics
78. NeuraStasis
79. LightLock Bio
80. BreakDown Bio, Inc.
81. Azalea Therapeutics
82. Acrigen Biosciences
83. Aikium
84. Ariz Precision Medicine
85. Bay Genomics LLC
86. Caribou Biosciences
87. Dextera Biosciences
88. Elenae Therapeutics
89. Inapill
90. Insamo
91. Iris Medicine
92. Izote Biosciences
93. Newomics
94. Novel Farms
95. Radar Therapeutics
96. Rareterra
97. Ray Therapeutics
98. ResVita Bio
99. Stork Therapeutics
100. Vivere Oncotherapies
101. Radar Therapeutics
102. Rareterra
103. Ray Therapeutics
104. ResVita Bio
105. Stork Therapeutics
106. Vivere Oncotherapies
107. Activate
108. Berkeley Air Monitoring Group, Inc

CLEANTECH

109. Calectra
110. Calion Technologies
111. Calyx
112. Capture6
113. Dauntless.io
114. Deep Fission
115. Enthalpy Analytical
116. Harvest
117. Huue Bio
118. HyWatts
119. Inlyte Energy
120. Iterant
121. Next-Ion
122. Noble Thermodynamics
123. Picoyune
124. PVComplete
125. SimpleLab
126. Treau
127. WaHa
128. Westwood Aerogel
129. Windscape AI
130. Winko Solar
131. Swift Charge
132. Transition Biomining
133. WattUp Energy
134. Deep Isolation
135. Form Energy
136. GridLab
137. KoBold Metals
138. OLIVINE INC
139. RareTerra Inc.
140. Serinus Labs, Inc.
141. RareTerra Inc.
142. AsparaGlue
143. Decoded Health
144. Imvaria
145. Millie

HEALTHCARE

146. Organos
147. Pelagius
148. Roger
149. Somnee
150. Digital Therapeutix, Inc.
151. Knowledge Synthesis
152. O.N. Diagnostics

HARDWARE

153. AutoMate Scientific
154. Bandwidth10
155. Bleximo
156. Foxeye Robotics
157. Hyperframe

158. Jitx
159. Luum
160. Mumec
161. NEYE Systems
162. OSO Semiconductor
163. Sensys Networks
164. Sonera Magnetics
165. Helping Robots Inc
166. Yektasonics
167. Maxivio
168. Ambi Robotics
169. Arris Composites
170. Avid Technology
171. PHOTONICS RESEARCH
172. Squishy Robotics
173. Zendar

OTHER

174. Kwikkart
175. Manara
176. BERKM
177. Edvisor.ai
178. AP0110
179. Quokka Prints
180. Ovidian Group
181. Reax Engineering, Inc.
182. Hardly Food

APPENDIX H: I&E Programs & Resources Distribution by Venture Development Stage

Ideation

1. Master of Engineering Program
2. Management, Entrepreneurship & Technology Program
3. Neurotech Collider Lab
4. Nucleate Bay Area
5. Ollie Legal
6. Office of Intellectual Property and Industry Research Alliances
7. PostX
8. Robinson Life Science, Business, and Entrepreneurship Program
9. she256
10. SkyDeck ACE Intern Program
11. Sutardja Center for Entrepreneurship & Technology
12. Tech Transfer (IPIRA) DEI Page
13. UCB Startup Fair
14. Venture Capital Investment Competition
15. YC Co-Founder Matching @ SCET
16. EDGE in Tech Initiative at U.C.
17. Free Ventures Fellowship
18. Fung Institute for Engineering Leadership (Fung Fellowship)
19. Garwood Center for Corporate Innovation
20. Haas Venture Fellows
21. Institute for Business Innovation
22. Inclusive Innovation Equitable Entrepreneurship
23. Innovation & Entrepreneurship Council
24. Jacobs Institute for Design Innovation
25. Learn2Launch
26. Decode
27. Design Scholars Program
28. Cradle to Commerce
29. CITRIS Invention Lab
30. Collider Cup
31. Cal Hacks
32. Cleantech to Market
33. Career Center
34. Berkeley Changemaker®
35. Berkeley Deep Tech Innovation Lab
36. Berkeley Discovery
37. Berkeley Entrepreneurs Association
38. Berkeley Haas Entrepreneurship Hub
39. Berkeley Haas Open Innovation Squad
40. Berkeley Venture Capital
41. BioEnginuity Impact Grant
42. Biology Scholars Program
43. Berkeley Haas Entrepreneurship Program
44. Berkeley Master of Design
45. Berkeley Postdoc Entrepreneurs Program

Early Formation

46. Master of Translational Medicine
47. QB3 Innovation Discovery
48. SCET – Summer Venture Lab
49. Startup@Berkeley Law
50. SBIR/STTR Grants
51. StEP
52. Tory Burch Fellowship

53. Energy and Biosciences Institute Entrepreneurial Business Incubator (EBI2)
54. FORM+FUND – Legal Support
55. Free Ventures
56. Health Engine
57. Haas Impact Fund
58. Handshake
59. NSF I-Corps Hub
60. Life Sciences Entrepreneurship Center
61. Decode
62. Dorm Room Fund
63. CITRIS Foundry
64. Cal Hacks Fellowship
65. Cyclotron Road
66. Bakar Labs
67. Berkeley Climate Change Network
68. Berkeley Deep Tech Innovation Lab
69. Berkeley Haas Entrepreneurship Hub
70. Berkeley Haas Entrepreneurship Startup Seed Fund
71. Big Ideas Contest
72. Bakar Fellows

Acceleration

73. QB3 Garage@Berkeley
74. QB3 Mentoring
75. Richmond Field Station
76. Health Tech CoLab
77. LAUNCH
78. Cal Alumni Association
79. Berkeley IP Practicum
80. Berkeley Research Infrastructure Commons
81. Berkeley Startup Cluster
82. Biomolecular Nanotechnology Center
83. Blockchain Xcelerator
84. Berkeley SkyDeck

Scale

85. House Fund
86. Impact Science Ventures
87. Ironfire Ventures
88. Kapor Capital
89. Double Bottom Line Ventures
90. CalTestBed
91. Courtyard Ventures
92. California Innovation Fund
93. Berkeley-Taiwan Health Innovation Accelerator
94. Bee Partners
95. Better Ventures
96. Bow Capital
97. Berkeley Catalyst Fund (I&II)
98. Berkeley SkyDeck Fund (I&II)
99. Blue Bear Ventures
100. Berkeley Angel Network
101. Berkeley Founders Pledge
102. Berkeley Haas Alumni Relations